1 Exam Practice

Question 1.1 (Practice 1)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function with compact support. Let $E \in \mathbb{R}$ and $\eta > 0$. Compute,

$$\lim_{\eta \to 0^+} \operatorname{Im}\left[\int_{\mathbb{R}} \frac{f(x)}{x-(E+i\eta)}\,dx\right]$$

If $\eta > 0$ show that the function

$$E \to \operatorname{Im}\left[\int_{\mathbb{R}} \frac{f(x)}{x-(E+i\eta)}\,dx\right]$$

is smooth.

Proof. Note that the function in the integral is

$$\operatorname{Im}\left(\frac{f(x)}{x-(E+i\eta)}\right) = \operatorname{Im}\left(\frac{(x-E+i\eta)f(x)}{(x-E)^2+\eta^2}\right) = \frac{\eta f(x)}{(x-E)^2+\eta^2}$$

Since f is continuous with compact support, we have f is only nonzero on $[-N, N]$ and $|f(x)| \le M$ on this interval. We then get

$$\left|\operatorname{Im}\left[\int_{\mathbb{R}} \frac{f(x)}{x-(E+i\eta)}\,dx\right]\right| \le \int_{-N}^{N} \frac{\eta M dx}{(x-E)^2+\eta^2} \le \int_{-N}^{N} \frac{M dx}{\eta} = \frac{2NM}{\eta} \to 0 \text{ as } \eta \to 0^+$$

This proves the first part.

Let F be that second function. Then we have

$$F(E) = \int_{\mathbb{R}} \frac{\eta f(x)}{(x-E)^2+\eta^2} dx = f * g(E)$$

where $g(t) = \frac{\eta}{t^2+\eta^2}$. Note that g is smooth. It thus suffices to show that $F' = f * g'$ by repeated application of this. For any E, h, we have

$$\begin{aligned}\left|\frac{F(E+h)-F(E)}{h} - f * g'(E)\right| &= \left|\frac{1}{h}\int_{\mathbb{R}} \frac{\eta f(x)}{(x-E-h)^2+\eta^2}dx - \frac{1}{h}\int_{\mathbb{R}} \frac{\eta f(x)}{(x-E)^2+\eta^2}dx - \int_{\mathbb{R}} g'(E-x)f(x)dx\right| \\ &\le 2MN \sup_{x\in[-N,N]} \left|\frac{g(E-x+h)-g(E-x)}{h} - g'(E-x)\right| \to 0 \text{ as } h \to 0\end{aligned}$$

The latter limit is uniform since g is C^1. Hence F is smooth. □

Question 1.2 (Practice 2)**.** Show that a monotonic function $f : \mathbb{R} \to \mathbb{R}$ is Borel measurable.

Proof. Without loss of generality, suppose f is increasing. It suffices to show that for any $(a, b) \subseteq \mathbb{R}$, $f^{-1}(a, b)$ is Borel measurable. If it is empty or a singleton then we are done. Otherwise, we show that $f^{-1}(a, b)$ is an interval. For that, we let $x < y \in f^{-1}(a, b)$ and show that any $t \in (x, y)$ is in $f^{-1}(a, b)$. Since f is increasing, we have $f(x) \le f(t) \le f(y)$, so $f(t) \in [f(x), f(y)] \subseteq (a, b)$, and hence $t \in f^{-1}(a, b)$. Therefore, f is Borel measurable. □

Question 1.3 (Practice 3)**.** If $m(E) = 0$ what can you say about $L^p(E)$?

Proof. Note that any set of outer measure zero is measurable, so any function $E \to \overline{\mathbb{R}}$ is measurable. Furthermore, for any such f, we have

$$\int_E |f|^p = 0 \Rightarrow f = 0$$

This shows that $L^p(E)$ contains only the zero function (under the equivalence relation). □

Question 1.4 (Practice 4)**.** For which p are trigonometric polynomials dense in $L^p(-\pi, \pi)$?

Proof. Recall that functions in $L^p(-\pi, \pi)$ can be approximated by continuous functions on $(-\pi, \pi)$ of bounded support. If we can show that each continuous function on $[a, b] \subseteq (-\pi, \pi)$ can be approximated by trigonometric polynomials, then we will have our result. One can use a Stone-Weierstrass Theorem argument to show that this can be done with the uniform convergence topology, so we just need to show that uniform convergence on $[a, b]$ implies L^p convergence on $(-\pi, \pi)$, given that the functions are zero a.e. outside $[a, b]$. Suppose conitnuous $f_n : (-\pi, \pi) \to \overline{\mathbb{R}}$ converge to a continuous f uniformly on $[a, b]$ and f, f_n are supported in $[a, b]$. Then, we have

$$\int_{(-\pi,\pi)} |f_n - f|^p \leq (b - a) \sup_{x \in [a,b]} |f_n(x) - f(x)| \to 0$$

as $n \to \infty$, which completes the proof. $\square$

Question 1.5 (Practice 5)**.** Let (M, d) be a metric space and for $x \in M$ and a nonempty set A define

$$d(x, A) := \inf_{y \in A} d(x, y)$$

Prove that $d(x, A)$ is a continuous function of x. Prove that A is closed iff the following statement holds: for all $x \in M$ we have that $d(x, A) = 0$ iff $x \in A$.

Proof. Suppose A is closed. Clearly if $x \in A$, we have $d(x, A) = 0$. If $x \notin A$, then since A is closed, x is an interior point of the open set $M \setminus A$, so there is some $r > 0$ such that $B_x(r) \subseteq M \setminus A$. This means that every point $y \in A$ satisfies $d(x, y) \geq r$, hence $d(x, A) \geq r > 0$.

Conversely, suppose that $d(x, A) = 0$ iff $x \in A$. We show that A is closed by showing that its complement is open. For any $x \in M \setminus A$, we let $r = d(x, A) > 0$. For any $y \in B_x(r)$, we have $d(x, y) < r = d(x, A) = \inf_{z \in A} d(x, z)$, so y cannot be in A. Hence, $y \in M \setminus A \Rightarrow B_x(r) \subseteq M \setminus A$, and therefore, A is closed. $\square$

Question 1.6 (Practice 6)**.** Let $f_n(x)$ be a sequence of polynomials and let $f(x) = \lim_{n\to\infty} f_n(x)$. Is f necessarily continuous?

Proof. Not necessarily. Consider the space $C[0, 1]$ and the polynomials $f_n(x) = x^n$. For $x \in [0, 1)$, $f(x) = \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} x^n = 0$, and $f(1) = \lim_{n\to\infty} 1 = 1$. Clearly, f is not continuous. $\square$

Question 1.7 (Practice 7)**.** Let (M, d) be metric space. We say a sequence $x_n \in M$ is fast-Cauchy if there is a summable sequence $a_n > 0$ (i.e., $\sum a_n < \infty$) such that $d(x_n, x_{n+1}) < a_n$. Disprove or prove the following:

(a) A fast-Cauchy sequence is Cauchy

(b) Every Cauchy sequence has a fast-Cauchy subsequence

(c) A sequence is Cauchy iff every subsequence has a subsequence that is fast-Cauchy

Proof. $\square$